Differential Gene Expression Analysis Of Lung Adenocarcinoma vs Normal Tissue Using TCGA RNA-seq Data

➢ Background

Lung Adenocarcinoma (LUAD) is one of the most prevalent histological subtypes, comprising around 40% of non-small cell lung cancer cases (NSCLC), and remains the leading cause of cancer-related global mortality.[1]. The significant inter-patient and intra-tumor heterogeneity poses a major clinical barrier in LUAD management[2].The genomic alterations in oncogenic drivers such as EGFR, KRAS, and TP53 are well characterized and TCGA profiling has demonstrated that RAS/RAF pathway is frequently modified[3].However, the individual-level mutations do not adequately capture the wider transcriptome dysregulation reported across patient populations [4]. Resolving this heterogeneity requires functional analysis at a larger scale.To address this gap, the current study compares 539 tumor samples with 59 solid tissue normal samples using publicly accessible RNA-sequencing data from TCGA-LUAD, DESeq2, clusterProfiler for Gene Ontology enrichment, STRING and Cytoscape for protein-protein interaction networks, and Kaplan–Meier analysis for survival interactions were used to estimate differential gene expression. When combined, these methods enable a methodical assessment of changes in global expression and aid in the identification of exploratory prognostic prospects.

➢ Method

1. Data acquisition and Processing

The publicly available RNA-sequencing count data for lung adenocarcinoma (TCGA-LUAD) were obtained from The Cancer Genome Atlas through the Genomic Data Commons (GDC) portal, with the help of TCGAbiolinks package in R.. The dataset comprised 539 primary tumor samples and 59 solid tissue normal samples.Prior to statistical modeling, low-count genes were eliminated by retaining only transcripts with a total read count of at least 10, which reduces noise from genes with too little signal to analyze reliably**.**

2. Differential Gene Expression Analysis

DESeq2 in R was used to examine differential gene expression between primary tumor and normal solid tissue samples. The genes were considered substantially differentially expressed if the adjusted p-value was less than 0.05 (Benjamini-Hochberg correction) and the absolute log2 fold change was more than one. A positive log2 fold change indicated upregulation in the tumor relative to normal tissue, and negative values indicated downregulation. The visualized results generated a volcano plot in EnhancedVolcano that shows statistical significance against effect size. A heatmap of the 50 significant genes with the lowest adjusted p-values was produced with pheatmap.

3. Functional Enrichment Analysis

Gene Ontology (GO) Biological Process enrichment analysis was conducted using clusterProfiler and the human annotation package org.Hs.eg.db to investigate the biological functions linked to upregulated genes in LUAD. Ensembl gene identifiers were converted to Entrez IDs, and terms with an adjusted p-value of less than 0.05 (after Benjamini-Hochberg correction) were considered highly enriched. A dot plot was used to display the most enriched biological processes consisting of gene ratios, significance levels, and gene count.The obtained results were used to identify biological processes that may be relevant to LUAD

4. Survival Analysis

Overall survival (OS) was evaluated using the survival and survminer R packages to observe whether gene expression is associated with patient outcomes. The GDC provided clinical data for TCGA-LUAD, and OS time was defined as days to death for deceased patients and days to last follow-up for live patients. The median DESeq2-normalized expression of SACK1A (also known as FAM83A) was used to distinguish patients into high- and low-expression groups after primary tumor samples were matched to clinical records. Kaplan–Meier survival curves were generated using the survminer package, while the survival package was used for survival analysis and the log-rank test Using the log-rank test, survival curves were compared. There is no independent predictive value established by this exploratory investigation.

5. Protein-Protein Interaction (PPI) Network Analysis

PPI analysis was performed by ranking significantly upregulated genes (adjusted p-value < 0.05, log2 fold change > 1) by adjusted p-value. The resultant top 100 genes were submitted to the STRING database for retrieval of known and predicted protein associations, using a high-confidence interaction score of ≥ 0.700. The obtained network was imported into Cytoscape for visualization and manual layout.

➢ Results

1. Differential Gene Expression Analysis

DESeq2 identified 15,156 notably differentially expressed genes (adjusted p-value < 0.05, |log2 fold change| > 1) between 539 primary tumor samples and 59 solid tissue normal samples. In comparison to normal tissue, 11,868 (78.3%) of these were upregulated and 3,288 (21.7%) were downregulated in tumors.Each gene's fold change and adjusted p-value are shown in the volcano plot (Figure 1). The significant gene changes can be observed in both directions, with more on the upregulated side, with fold changes ranging from -6.2 to +10.5. The key significant genes, FAM83A, PYCR1, and OTUD1, appear at the top of the plot.The heatmap presented consists of the 50 most significant genes (Figure 2) where most normal samples are distinguished from the tumor samples, while some samples clustered with the other group.The genes have been classified into two categories: 29 had higher expression in tumors and 21 were more highly expressed in normal tissue. Contrary to normal samples, expression in tumor samples indicated to be more varied.

Table 1 displays the five most significant genes in order of adjusted p-value. In tumor tissues, OTUD1 and EPAS1 were downregulated while SACK1A(FAM83A), PYCR1, and AFAP1-AS1 were elevated.These findings exhibit considerable transcriptional differences between normal lung tissue and LUAD tumors.

Gene	Ensemble ID	Direction	Log 2 Fold Change	Adjusted P value
SACK1A(FAM83A)	ENSG00000147689.16	Up	6.79	4.3×10^{-176}
PYCR1	ENSG00000183010.17	Up	3.69	3.0×10^{-165}
OTUD1	ENSG00000165312.6	Down	-2.12	9.4×10^{-146}
AFAP1-AS1	ENSG00000272620.2	Up	6.31	1.2×10^{-128}
EPAS1	ENSG00000116016.14	Down	-2.72	4.2×10^{-128}

Table 1 : Top five differentially expressed genes ranked by adjusted p-value

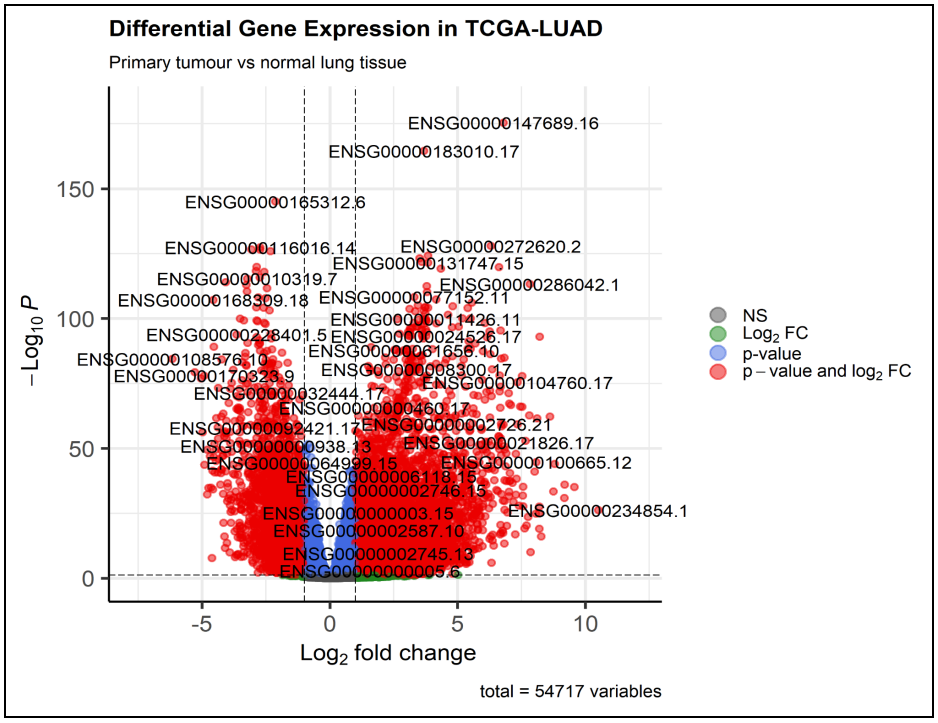


Figure 1 : Volcano plot of differential gene expression between primary tumor and solid tissue normal samples in TCGA-LUAD. Each point is one gene. The x axis indicates log 2 fold change (tumor vs normal) and the y axis stands for -log 10 of adjusted p value. Points not meeting the threshold are shown in gray. Red points meet both thresholds, Green points meet only the log 2 fold change threshold and blue points meet only p value threshold. While the dashed lines mark the thresholds (|log2 fold change| = 1, adjusted p-value = 0.05).

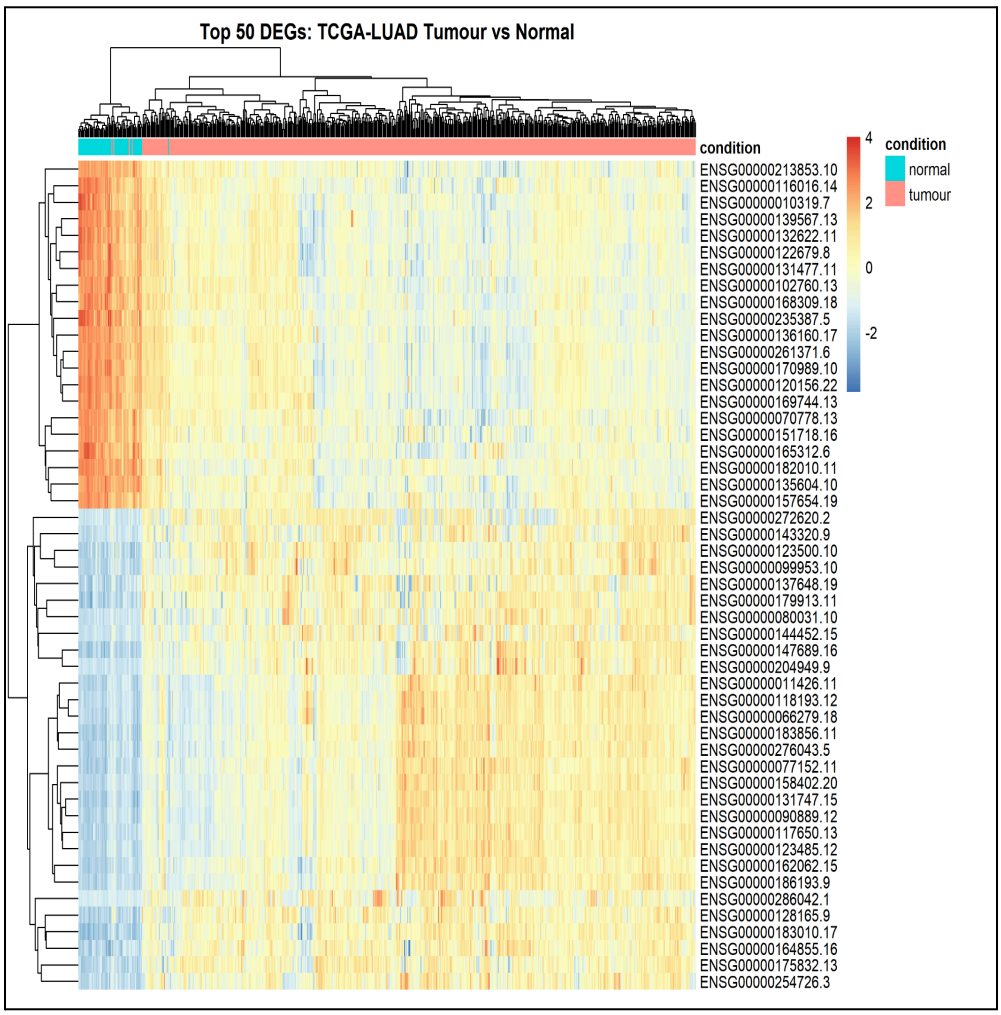


Figure 2: The given figure presents a heatmap of the 50 most significantly expressed genes, ranked by adjusted p value. The rows are genes and columns are samples that are annotated by tissue type. Red indicates expression above a gene's average across sample and blue signifies below average. Genes and samples with similar patterns are grouped together.

2. Functional Enrichment

The enrichment of upregulated genes in Gene Ontology Biological Process revealed 237 significant terms (adjusted p-value < 0.05; Figure 3). The resultant dot plot visualized consists of dots representing individual GO terms, where its horizontal position shows the proportion of analyzed genes in the term, size indicates the number of genes and its color shows the adjusted p-value. In line with the consistent proliferation expected in LUAD, cell division and chromosome segregation were observed in the top terms, including organelle fission (143 genes), nuclear division (138 genes), and nuclear chromosome segregation (98 genes). Chromatin terms such as nucleosome assembly were also enriched and consisted largely of histone genes. Immune processes like B cell-mediated immunity, were driven majorly by immunoglobulin genes, which may show the presence of immune cells in the tumor microenvironment. Processes that were less expected included meiotic processes, which share most of their genes with nuclear division, and skeletal system development (107 genes), which this analysis does not explain. This enrichment demonstrates overrepresentation of genes in these processes, not that the processes are activated.

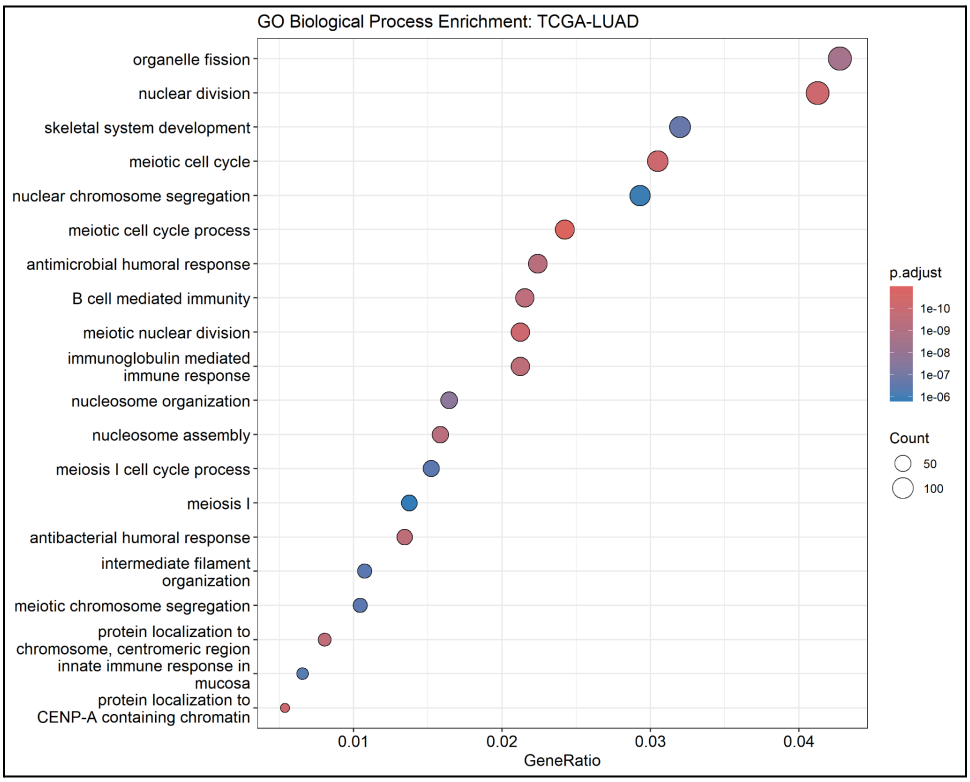


Figure 3: GO Biological Process enrichment of genes upregulated in TCGA-LUAD tumors. The plot showcases the 20 terms (biological processes) with the highest gene ratios. The x-axis indicates gene ratio, dot size shows gene count, and color shows the adjusted p-value (red = stronger significance).

3. Survival Analysis

To see whether the activity of the most significantly upregulated gene, SACK1A (also known as FAM83A) is linked to overall survival of the patients with LUAD, 504 TCGA-LUAD patients were categorized into two equal groups of 252 based on the median expression of this gene (3,507.5 normalized counts): a low-expression group and a high-expression group. Survival was then compared using a Kaplan-Meier plot, which shows the count of patients still alive over time, and a log-rank test, which checks whether the difference between two such curves is larger than chance would explain .Patients with high SACK1A expression had a poor outcome. Deaths were observed in 106 patients (42.1%) in the high-expression group, compared with 76 (30.2%) in the low-expression group. As shown in the plot, the high-expression curve is found to be below the low-expression curve for most of the follow-up period..The predicted

median survival time was 3.5 years in the high-expression group and 4.5 years in the low-expression group. The difference was statistically significant (log-rank $\chi^2 = 11.65$, $p = 0.000642$). Estimates beyond approximately 2,000 days are less clear, as few patients were still being tracked at that point. This result indicates an association between higher SACK1A expression and poorer survival (higher mortality rate). However, it does not determine that SACK1A causes worse outcomes, and factors such as age and tumor stage were not taken into consideration.

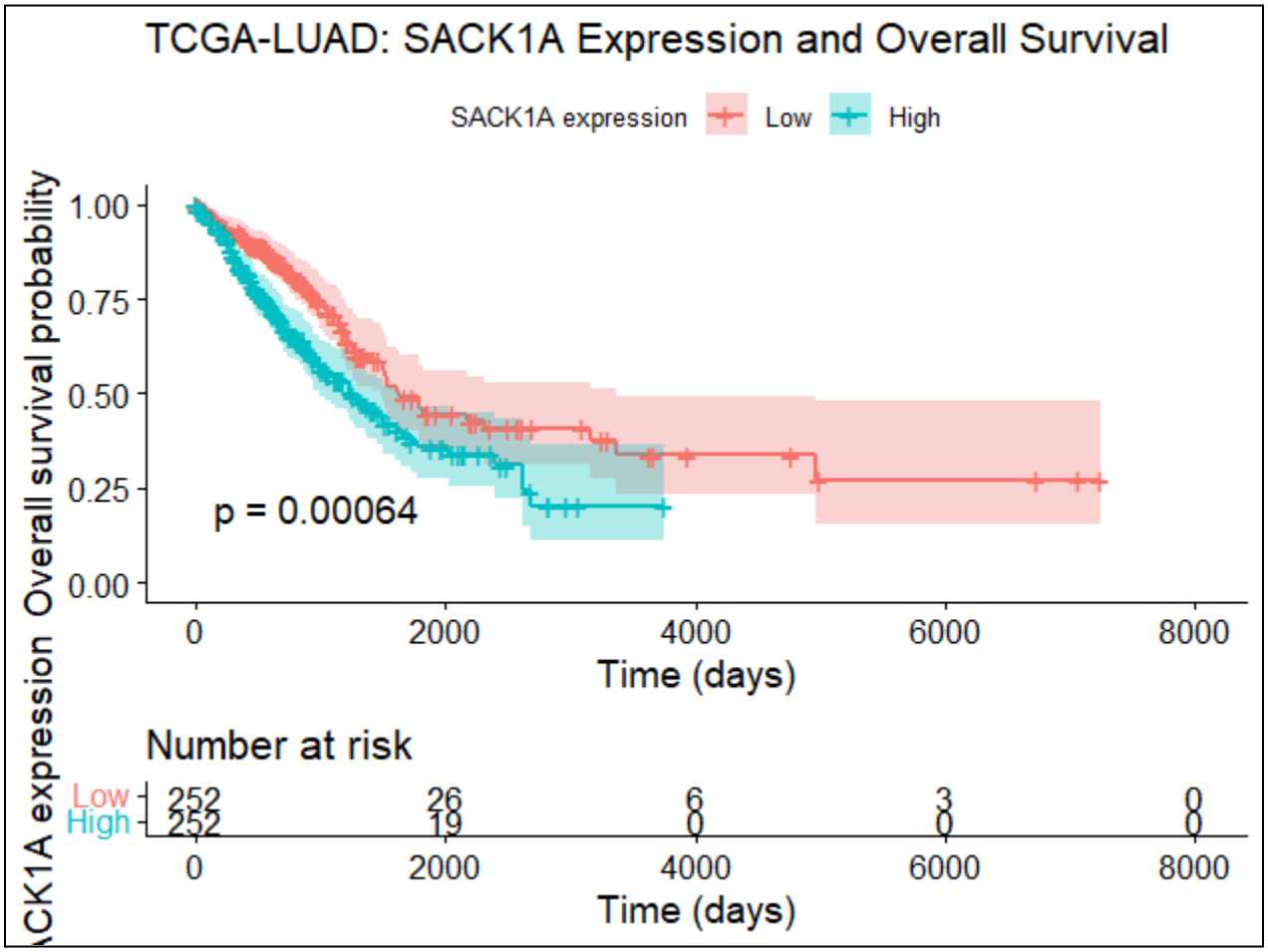


Figure 4: Kaplan-Meier plot of TCGA-LUAD patients' overall survival based on SACK1A expression. The shaded band displays the 95% confidence interval, and each line represents the percentage of patients who have survived throughout time The table below shows how many patients were still being tracked at each time point. The p-value is from the log rank test

4. Protein-Protein Interaction Network

The 100 most significant upregulated genes (arranged by adjusted p-value) were uploaded to STRING, a database of known and predicted protein interactions, retaining only high-confidence linkages (combined score ≥ 0.700).

STRING network (Figure 5A). STRING recognized 90 of the 100 genes that were submitted. Of these, 55 established at least one high-confidence interaction, while the rest 35, including FAM83A, had none at this level.

Cytoscape network (Figure 5B). The 55 connected genes formed 505 interactions, 50 of which joined a densely connected cluster. The most linked genes (hubs) were DLGAP5 (36 connections), BUB1B and TOP2A (35 each), CCNB1 and KIF2C (34 each), and CDC20 (33). These are mitosis and cell division genes, consistent with the cell-cycle terms in GO enrichment results. Smaller groups that lay outside the cluster include (COL10A1, COL11A1, and MMP11; FUT2 and B3GNT3), and PYCR1 connected to the network only through ALDH18A1. Connectivity alone does not imply that a gene is functionally relevant or causal.

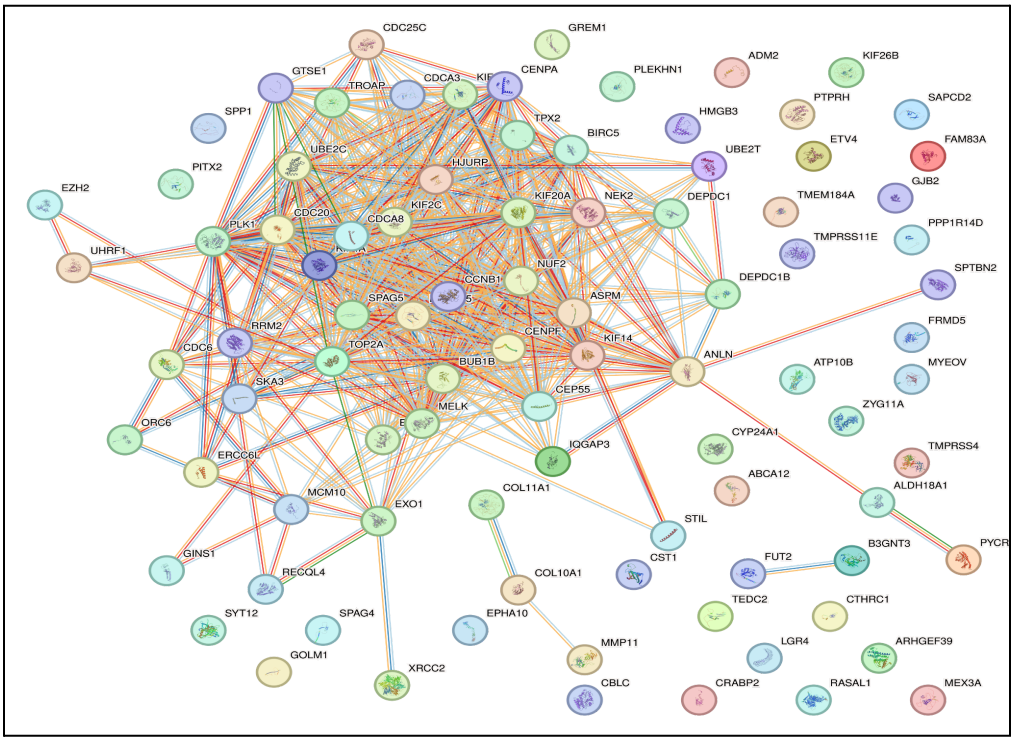


Figure 5A. STRING protein-protein interaction network of the 100 most significant upregulated genes in TCGA-LUAD (90 nodes mapped; minimum combined score ≥ 0.700). Each circle is a gene, and the colors of the lines show the kind of evidence behind each interaction. Genes shown on their own, including FAM83A, had no interactions at this confidence score.

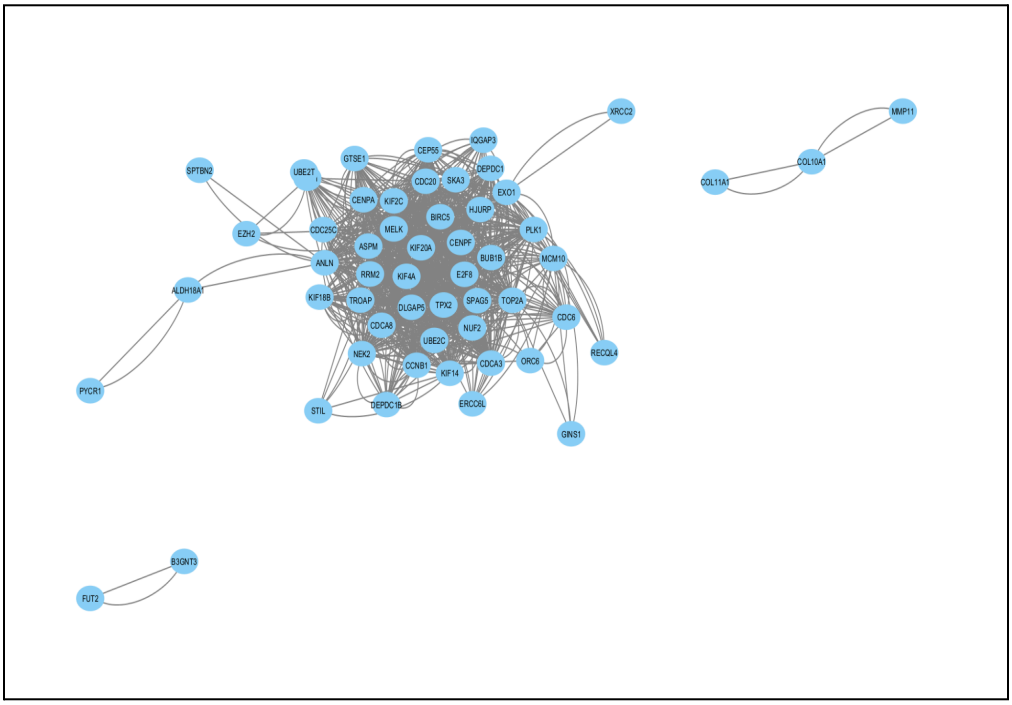


Figure 5B. Cytoscape visualization of the 55 nodes with at least one interaction (505 unique edges; combined score ≥ 0.700), arranged in a manual layout. The main component contains 50 nodes, with the highest-degree nodes DLGAP5, BUB1B, TOP2A, CCNB1, KIF2C, and CDC20.

➢ Discussion

This study involved comparison of 539 primary LUAD tumors with 59 normal lung samples and found widespread expression changes, with 15,156 differentially expressed genes, 78.3% of them upregulated. The signal that predominated was proliferation.The most connected genes in the interaction network (DLGAP5, BUB1B, TOP2A, CCNB1, KIF2C, and CDC20) are all associated with nuclear division, organelle fission, and chromosome segregation terms. Several of these genes, including CDC20, BUB1B, and DLGAP5, have been identified as hub genes in other LUAD studies, and in another LUAD analysis, high expression of genes such as TOP2A, CDC20, and BUB1B was linked to shorter overall survival. [5,6]

The enrichment results seem rational for this cancer type, because cell division mechanisms are also documented among upregulated genes in NSCLC.[7] The traditional LUAD driver genes EGFR, KRAS, and TP53 did not meet the |log2 fold change| > 1 criteria since they had small fold changes (log2 fold changes 0.41, 0.64, and 0.52) but were statistically higher in tumors (adjusted p-values 0.029, 6.6×10^{-9} and 2.1×10^{-6}). A lack of significant expression changes is not surprising because these genes are mostly modified by mutation. The lung epithelial genes SFTPC and AGER were significantly downregulated (-4.74 and -5.04), which may indicate less normal alveolar cells in tumor tissue, while the proliferation marker MKI67 was increased (log2 fold change 3.10).[8,9,10]

The top five genes support previous findings. The most crucial gene, FAM83A (SACK1A), relies on EGFR signaling for its expression, is overexpressed in lung cancer, and is associated with poor survival [16]. While OTUD1 and EPAS1 are known to be downregulated in lung cancer, with OTUD1 functioning as a tumor suppressor [21] and EPAS1 being silenced by promoter hypermethylation [14], PYCR1 is linked to suppression of antitumor immunity and AFAP1-AS1 a supposed predictor of overall survival ,both are reported to be elevated [12, 20]. Expression variations by itself are unable to determine whether these genes cause tumor growth or simply reflect it. While the loss of EPAS1 is more likely to be the result of epigenetic silencing [14], experiments decreasing FAM83A or AFAP1-AS1 suggest they may contribute to tumor cell development [13,16], and PYCR1 has been connected to tumor progression [20]. Establishing causality would require functional experiments.The growth related roles reported for FAM83A, PYCR1, and AFAP1-AS1 [13,16,20] fit the proliferative signal in the GO plot, even though these genes are not themselves part of the enriched cell division terms.

The three analyses support each other for the cell-cycle signal but not for every top gene. Nuclear division comprises 23 of the 55 associated genes, and all six network hubs fall under the categories of nuclear chromosomal segregation, organelle fission, and nuclear division.The coordinated upregulation of the six hub genes fits a tumor consisting of dividing cells, although this is an interpretation on the basis of expression patterns and not a direct measurement of cell division .PYCR1 and ALDH18A1, which are linked in the network, are both annotated to amino acid metabolic process terms in the entire enrichment table, and COL10A1 and COL11A1, grouped separately with MMP11, include genes behind the skeletal system development term. The strongest single-gene signal and the dominant pathway signal are distinct lines of evidence because FAM83A is not annotated to any enriched term and formed no link at the 0.700 threshold. AFAP1-AS1 is non-coding and so cannot appear in a protein network, although EZH2, which it has been reported to interact with in NSCLC, does appear [12]. Since only upregulated genes were included, OTUD1 and EPAS1 were intentionally left out of the network.

Several findings were less expected.The meiotic terms are most likely explained by shared genes, as 78% to 100% of their genes are involved in nuclear division, and the network's meiotic genes, including CDC20, TOP2A, and NUF2, are well-known mitotic genes[11]. Collagen, HOX, and DLX family genes were involved in skeletal system

development, and their overexpression could be due to extracellular matrix or developmental programs, a concept that requires further investigation. . The immune terms were majorly driven by immunoglobulin genes, implying B cells or plasma cells in the tumor, whereas the mucosal and antimicrobial terms were primarily driven by histone genes, and should not be interpreted as immune activity.

Patients with high FAM83A expression had severe overall survival (106 vs. 76 deaths; log-rank $p = 0.000642$), in line with published findings. FAM83A overexpression has been linked to early lymph node metastasis and poor overall survival in LUAD.[18] The result presented here is an association only. Because FAM83A is tied to EGFR signaling, tumor stage and EGFR mutation status could partly explain the difference, and none of these factors, nor treatment, was adjusted for [17]. Previous TCGA-based work identified FAM83A as an independent prognostic factor in a multivariable analysis (adjusted hazard ratio 2.17), but because it used TCGA data, agreement with it does not constitute independent replication. [16]

Several limitations apply. As the comparison is unbalanced (539 tumor vs. 59 normal samples), the variance estimates for the smaller group may be less accurate. TCGA data were generated across many centers and over time, so batch effects and technical variation (sequencing depth, RNA quality, library preparation) may influence the results. Because the Bulk RNA-seq consists of a mix of tumor, immune, and stromal cells, some cell type specific signals, such as the immunoglobulin genes from infiltrating immune cells, cannot be attributed to tumor cells themselves. With a total of 598 samples, the analysis had high statistical power ,and the low-count filter was permissive, so several genes passed the thresholds, and not all of them are necessarily biologically important..GO enrichment output included only upregulated genes with annotations (3,344 genes), the network used only the top 100 upregulated genes, and STRING favors well-studied proteins. The survival analysis applied a median split, did not adjust for clinical factors, and had few patients at risk after 2,000 days, making estimates in that range less certain. The treatment history of the patient was not examined.

Since the survival results so far are dependent on a single dataset, the most critical next step is to determine if these findings hold in an independent cohort of LUAD patients. A more extensive survival analysis might then be performed to see whether FAM83A predicts outcome on its own, after taking into consideration factors such as tumor stage, age, smoking history, EGFR and KRAS mutation status, and treatment. It would also be useful to investigate whether the top genes become more active in later stages of disease, indicating a link to tumor progression rather than simply a difference between tumor and normal tissue. Because bulk sequencing combines signals from multiple cell types, single-cell data or computational estimations of cell composition may reveal which signals come from tumor cells and which from surrounding immune and support cells. The network analysis might also be expanded to include downregulated genes like OTUD1 and EPAS1, which were skipped out here. Finally, laboratory investigations that inhibit FAM83A, PYCR1, or AFAP1-AS1 in lung cancer cell lines would determine whether these genes contribute to tumor growth. Overall, the findings indicate a significantly proliferative expression profile in LUAD, with FAM83A emerging as a promising prospective marker, and should be interpreted as a lead for a research study rather than established conclusions.

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